

Join

- Join is a special form of cross product of two tables.
- In Cartesian product we join a tuple of one table with the tuples of the second table. But in join there is a special requirement of relationship between tuples.
- For example if there is a relation STUDENT and a relation BOOK then it may be required to know that how many books have been issued to any particular student. Now in this case the primary key of STUDENT that is stId is a foreign key in BOOK table through which the join can be made.

SQL Joins

Introduction:-

An SQL JOIN clause combines rows from two or more tables. It creates a set of rows in a temporary table.

Types of join

- Inner Join
- Outer Join
- Cross Join/ Cartesian Join

CROSS JOIN

It is also known as **CARTESIAN JOIN**, which returns the Cartesian product of two or more joined tables. The **CROSS JOIN** produces a table that merges each row from the first table with each second table row. It is not required to include any condition in CROSS JOIN.

Syntax:

```
Select * from table_1 cross join table_2;
```

Or,

```
Select column1, column2, column3 FROM table_1, table_2;
```

Table 1: Product_Details

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ID	ProductName	Amount
Pro101	Laptop	56000
Pro102	Mobile	38000
Pro103	Headphones	5000
Pro104	Television	25000
Pro105	iPad	60000

Table 2: Customer_Details

CustomerName	CustomerAddress	CustomerAge	ProductID
Martin Guptill	San Francisco, USA	26	Pro101
James	Australia	29	Pro103
Ambati Williamson	New Zealand	27	Pro102
Jofra Archer	South Africa	24	Pro105
Omen	England	29	Pro107
Morgan	England	20	Pro108

Select ID, ProductName, CustomerName, CustomerAddress, Amount FROM Product_Details, Customer_Details;

After executing the query, it produces the below table.

ID	ProductName	CustomerName	CustomerAddress	Amount
Pro101	Laptop	Martin Guptill	San Francisco, USA	56000
Pro102	Mobile	Martin Guptill	San Francisco, USA	38000
Pro103	HeadPhone	Martin Guptill	San Francisco, USA	5000
Pro104	Television	Martin Guptill	San Francisco, USA	25000
Pro105	iPad	Martin Guptill	San Francisco, USA	60000
Pro101	Laptop	James	Australia	56000
Pro102	Mobile	James	Australia	38000
Pro103	HeadPhone	James	Australia	5000
Pro104	Television	James	Australia	25000
Pro105	iPad	James	Australia	60000
Pro101	Laptop	Ambati Williamson	New Zealand	56000
Pro102	Mobile	Ambati Williamson	New Zealand	38000
Pro103	HeadPhone	Ambati Williamson	New Zealand	5000
Pro104	Television	Ambati Williamson	New Zealand	25000
Pro105	iPad	Ambati Williamson	New Zealand	60000
Pro101	Laptop	Jofra Archer	South Africa	56000
Pro102	Mobile	Jofra Archer	South Africa	38000
Pro103	HeadPhone	Jofra Archer	South Africa	5000
Pro104	Television	Jofra Archer	South Africa	25000
Pro105	iPad	Jofra Archer	South Africa	60000
Pro101	Laptop	Omen	England	56000
Pro102	Mobile	Omen	England	38000
Pro103	HeadPhone	Omen	England	5000
Pro104	Television	Omen	England	25000
Pro105	iPad	Omen	England	60000
Pro101	Laptop	Morgan	England	56000
Pro102	Mobile	Morgan	England	38000
Pro103	HeadPhone	Morgan	England	5000
Pro104	Television	Morgan	England	25000
Pro105	iPad	Morgan	England	60000

30 rows in set (0.00 sec)

Inner Join

Will return those tuples which will be matching the condition

- Equi Join
- Natural Join

Equi-join

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An *equi-join* is a join based on equality or matching column values. This equality is indicated with an equal sign (=) as the comparison operator in the WHERE clause, as the following query shows.

Figure 1. Query

```
SELECT * FROM manufact, stock
WHERE manufact.manu_code = stock.manu_code;
```

SQL Equi Join

Introduction:-

SQL EQUI JOIN performs a JOIN against equality or matching column(s) values of the associated tables. An equal sign (=) is used as comparison operator in the where clause to refer equality.

You may also perform EQUI JOIN by using JOIN keyword followed by ON keyword and then specifying names of the columns along with their associated tables to check equality.

id	name	class	city
3	Hina	3	Delhi
4	Megha	2	Delhi
6	Gouri	2	Delhi

Table name — Record

In this table, you have l'd, class and city are the fields.

```
Select * from Record;
```

id	class	city
9	3	Delhi
10	2	Delhi
12	2	Delhi

then relate tables to check equality, using equal sign (=).

Syntax :

```
SELECT column_list
FROM table1, table2....
WHERE table1.column_name =
table2.column_name;
```

Example –

```
SELECT student.name, student.id, record.class, record.city
FROM student, record
WHERE student.city = record.city;
```

```
SELECT column_list
FROM table1
JOIN table2
[ON (join_condition)]
```

Example –

```
SELECT student.name, student.id, record.class, record.city
FROM student
JOIN record
ON student.city = record.city;
```

Output :

Input :

name	id	class	city
Hina	3	3	Delhi
Megha	4	3	Delhi
Gouri	6	3	Delhi
Hina	3	2	Delhi
Megha	4	2	Delhi
Gouri	6	2	Delhi
Hina	3	2	Delhi
Megha	4	2	Delhi
Gouri	6	2	Delhi

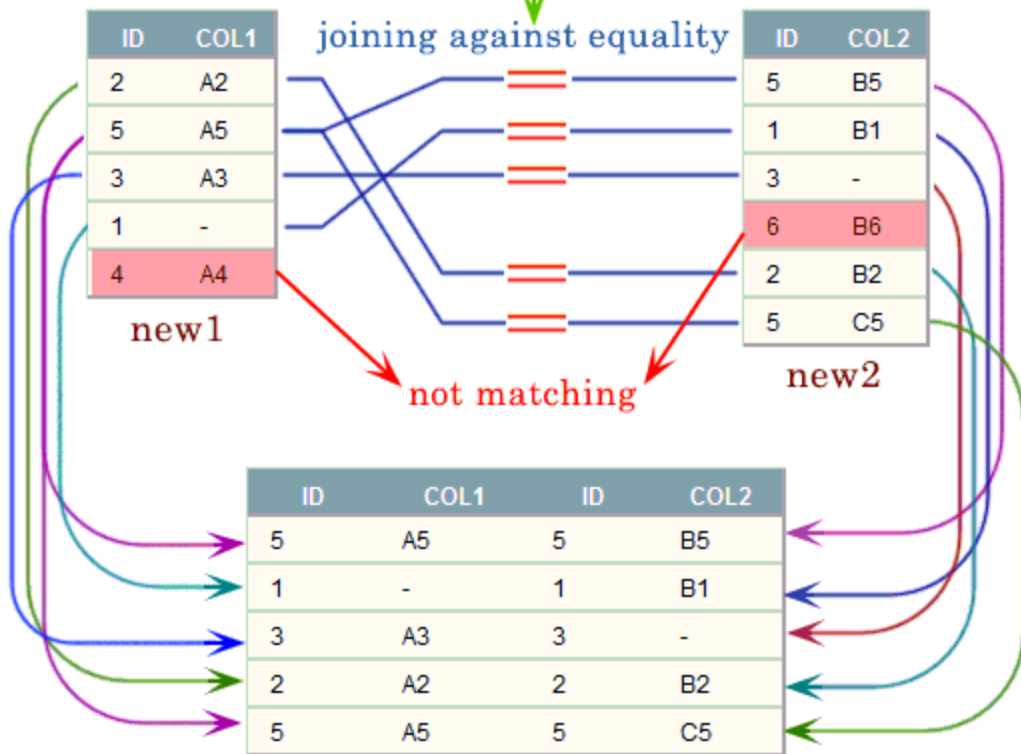
ID	COL1
2	A2
5	A5
3	A3
1	-
4	A4

new1

ID	COL2
5	B5
1	B1
3	-
6	B6
2	B2
5	C5

new2

```
SELECT * FROM new1
JOIN new2
ON new1.id = new2.id;
```

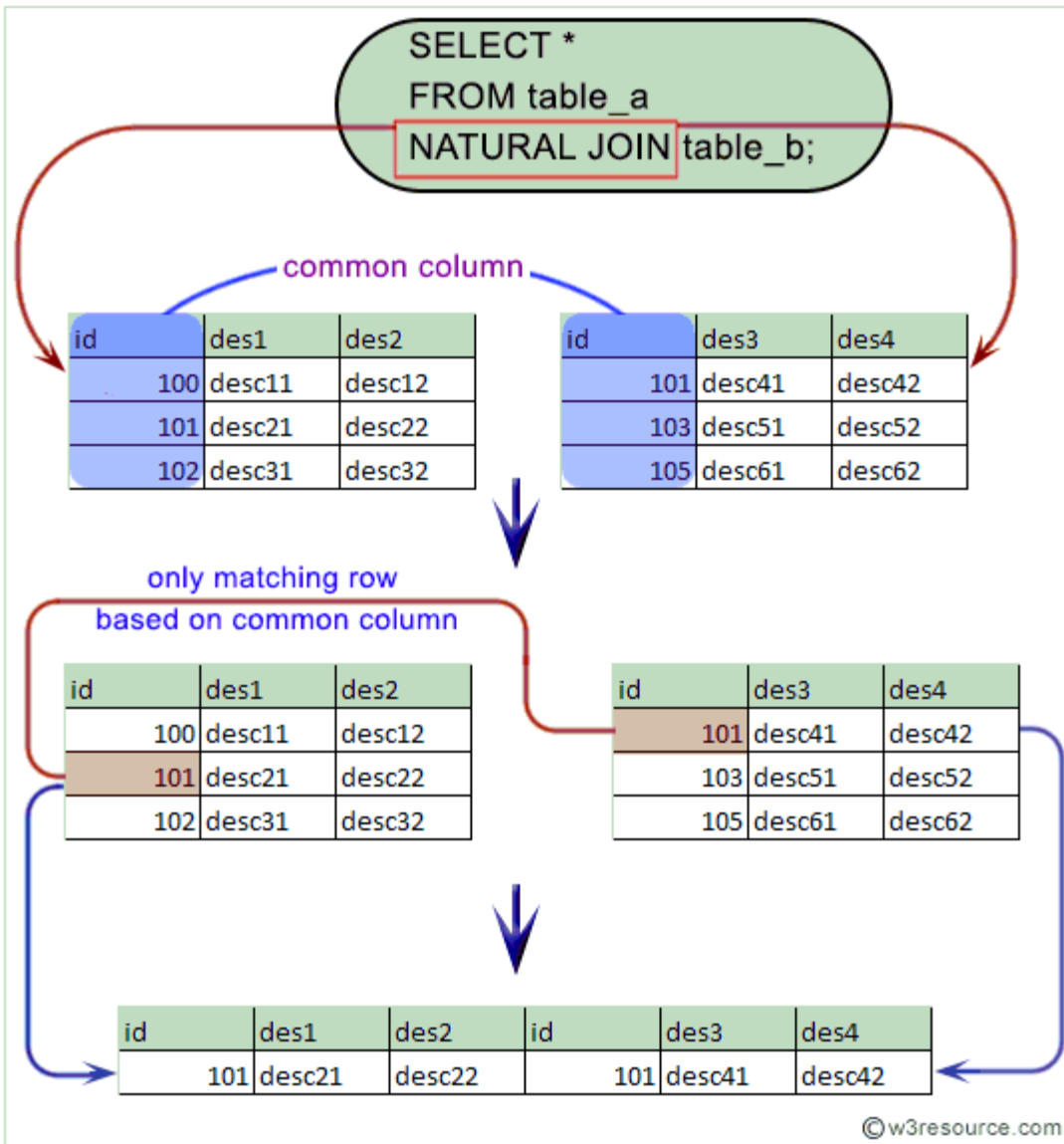


What is Natural Join in SQL?

We have already learned that an EQUI JOIN performs a JOIN against equality or matching column(s) values of the associated tables and an equal sign (=) is used as comparison operator in the where clause to refer equality.

The SQL NATURAL JOIN is a type of EQUI JOIN and is structured in such a way that, columns with the same name of associated tables will appear once only.

Pictorial presentation of the above SQL Natural Join:



Difference between natural join and equi join



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Natural Join automatically join two tables on columns which have equal name and in the result, the common column would appear one time , but in equi join we explicitly mention the condition that on which column we have to do equality test and also the columns would appear twice.



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